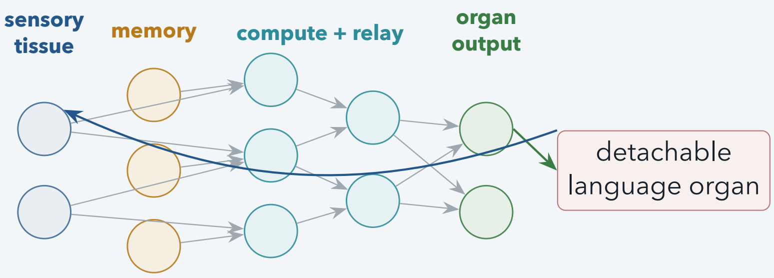
DMON-1 Research Project

Toward a Persistent, Growing Neural Organism

Typed cellular computation, detachable cognitive organs, continual adaptation, and frozen-LLM language control



Working research report

4 August 2026

This document reports implemented systems and completed experiments through L0-C1g. Claims are labeled by evidential status; exploratory results are not promoted into capability claims.

Abstract

DMON-1 is an attempt to build a persistent neural creature rather than a stateless task model: a continuously stimulated cellular network whose internal state, learned anatomy, developmental history, and detachable organs together constitute the agent. The current implementation, SOL2, uses typed neural tissues on a sparse directed connectome, bounded recurrent operators, private cell expression and low-rank private transitions, stream-written memory, dedicated output tissue, append-only growth, and complete living-state checkpoints. Language is treated as a replaceable organ. A frozen Llama-3 8B model supplies contextual sensory features and fluent decoding, while the organism alone owns new trainable state and emits bounded low-rank controls.

The program has established several mechanistic results. Cellular recurrent state is causal for character prediction; bounded typed tissue yields replicated dependence on cell identity and learned topology; a mastered organism accelerates acquisition through a newly attached organ; removed organs can be reattached and rapidly recovered; utility anchors substantially improve the stability–plasticity frontier; and a frozen 8B language model can be integrated with exact zero-control parity and zero backbone gradients. The program has not yet established simultaneous mastery of multiple deep procedures, useful recruitment of pressure-grown neurons, autonomous conversation, or held-out factual recall through the organism. The latest memory experiments localize a real signal from retained memory through internal tissue, but show collapse at output tissue and insufficient training of the LLM-control geometry. These results support the feasibility of the organism concept while sharply defining the remaining work.

Contents

1 Research Objective

The project asks whether an artificial neural system can be organized as a living, developing substrate rather than as a fixed inference graph. The long-term target is a small creature that can grow, remain continuously active, acquire new organs, integrate new experience into a persistent internal world, and converse with a user. "DMON" is an intentional contraction of "Digimon": the desired endpoint is not merely a better benchmark model, but a recognizable individual with whom interaction is possible.

This changes the role of language modelling. A large language model is treated as Broca-like machinery: a powerful organ for linguistic representation and production, not the whole cognitive organism. The cellular substrate is intended to provide the rest of the loop: lifetime memory, mode selection, cross-modal integration, adaptation, development, motivation, and continuity of self. The design therefore does not aim to beat a transformer or GRU on every small closed-world task. Those models remain useful controls, but low bits-per-character alone cannot measure growth, lifelong plasticity, or multi-organ integration.

Central feasibility question

Can a persistent, locally connected neural organism develop useful internal structure, retain and reuse computations while its interfaces change, add capacity without being rebuilt, and control detachable pretrained organs without surrendering its identity to those organs?

The project deliberately distinguishes four levels of evidence:

- **Established** means a software or numerical contract was directly verified, usually with matched controls and deterministic artifacts.
- **Mechanistic evidence** means a causal intervention localized behavior to cellular state, tissue, topology, an organ, or newly introduced machinery.
- **Diagnostic** means an experiment usefully localized a bottleneck but did not satisfy the capability gate.
- **Open** marks a project goal for which present evidence is insufficient.

2 Design Commitments

2.1 The creature is the network

Earlier work explored a visible morphology on a resource field. That work remains useful as a future display organ, but it is not the central creature. In the current architecture, the creature is the persistent neural network itself. Sensory modalities, language, audio, and visible morphology are organs coupled to the same cellular state. This distinction prevents a visually compelling body from being mistaken for the computational organism.

2.2 Persistent state and continual stimulation

The organism does not conceptually begin and end with task episodes. Its state survives tokens, optimizer boundaries, checkpoint/resume, organ removal, and organ reattachment. Current result-bearing trainers still use finite truncated backpropagation windows; therefore the implemented system should be described as a persistent stream learner, not yet as a fully asynchronous biological learner. A separate versioned runtime keeps foreground ticks and background optimization distinct, but the strict asynchronous milestone remains open.

2.3 Anatomy should matter, but lesions need not be tolerated

Input, memory, compute, relay, and output tissue have different roles. Output cells are sinks that read only compute and relay tissue, so a decoder cannot directly bypass the body. Acute freeze, state-shuffle, identity-shuffle, and topology-rewire experiments are used to identify what carries a learned capability. They are not adversarial conditions the organism is expected to survive. In biological terms, destroying grey-matter state or rewiring white-matter routes should damage the learned computation.

An important methodological correction emerged during the Fable experiments: *freezing* a population measures whether it does work, while *shuffling* within a tissue measures whether cell identity or placement is differentiated. Useful mean-field tissue can be nearly shuffle-invariant. Reporting both avoids calling an interchangeable population "unused."

2.4 Reserve is permitted

The project does not reward every neuron for being active. Some internal cells may be dormant or weakly used, providing reserve for later plasticity. The relevant conditions are that a nontrivial subset is causally useful, the active computation traverses the organism, and claimed new tissue is actually recruited when growth is invoked.

2.5 Information metabolism and satiety

The intended economy is denominated in reducible information, not raw activity or raw surprise. Paying for unpredictable input creates a noisy-TV optimum; paying for prediction progress rewards structure that can be learned. Metabolism is currently deferred from SOL2 until capability and continual credit assignment are sufficiently understood. The eventual human loop also requires satiety: a creature rewarded without bound for attention would be structurally encouraged to manipulate its user.

3 The SOL2 Organism

3.1 Typed tissues on a sparse directed connectome

SOL2 replaces the earlier accidental assumption of one universal cell rule with one rule per tissue and bounded private expression per mutable cell. The current tissue layout is:

- **Input tissue.** Receives bounded organ-specific sensory drive and begins recurrent integration.
- **Stream-memory tissue.** Receives differentiable external writes through a finite FIFO; mutable cells cannot write it directly.
- **Compute tissue.** Performs recurrent transformation under its typed rule and private expression.
- **Relay tissue.** Supplies transport capacity, append-only growth, and an anatomical reserve that can connect old and new circuitry.
- **Output tissue.** Is cleared at token boundaries, receives only compute/relay input, and is read by the active output organ.

Each target cell owns a fixed number of dendrite slots. Installed slots name explicit source cells; dormant slots carry exactly zero message and reserve anatomical capacity. Attention is local to those named dendrites. Output cells are never sources and their dendrites may not read input or memory tissue, which makes the no-bypass constraint executable rather than rhetorical.

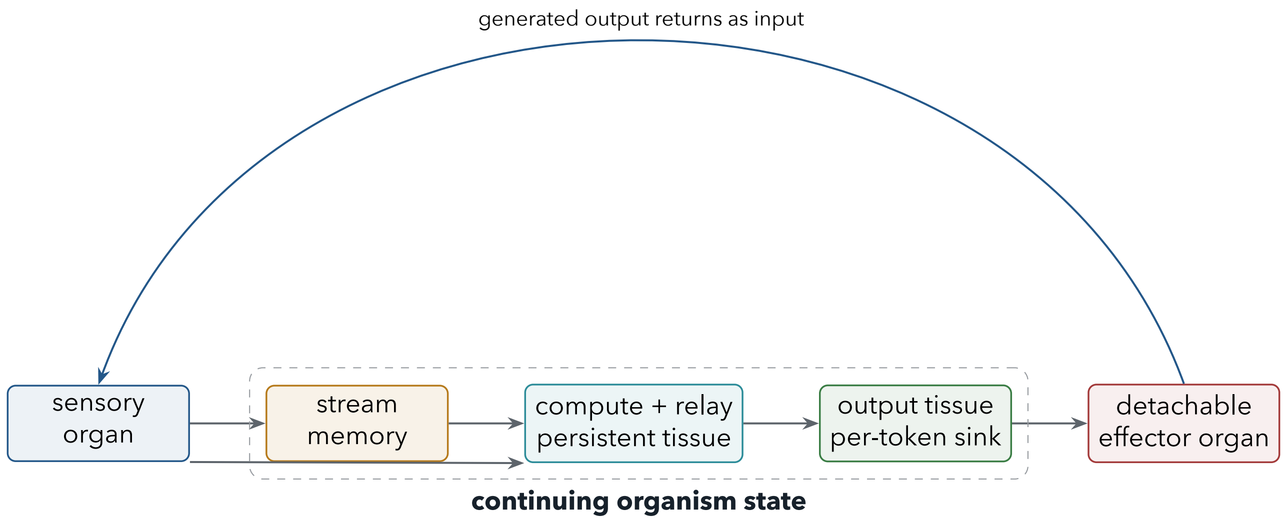


Figure 1: *The organism owns persistent state and recurrent anatomy. Sensory and effector organs may be attached or removed; all behaviorally relevant control must traverse compute/relay tissue and the dedicated output sink.*

3.2 Bounded cellular dynamics

For mutable cell i at transition t, SOL2 combines current state $h_{i,t}$, aggregated dendritic message $m_{i,t}$, and organ drive $d_{i,t}$. A typed tissue rule produces a bounded target, optionally shifted by private expression e_i and a private low-rank residual r_i:

$$z_{i,t} = f_{\tau(i)}([h_{i,t}, m_{i,t}, d_{i,t}]) + e_i + r_i, \tag{1}$$
$$\alpha_{i,t} \in [0.02, 0.80], \tag{2}$$
$$h_{i,t+1} = (1 - \alpha_{i,t})h_{i,t} + \alpha_{i,t} \tanh(z_{i,t}). \tag{3}$$

The bounded update keeps mutable state in $[-1, 1]$ from a zero initial state. Private target/time-constant expression gives cell identity without requiring a separate dense network per cell. Later experiments add rank-four private transition residuals whose up projection begins at zero, so the treatment is initially behaviorally silent and can be recruited by gradient descent.

For target i with installed sources $\mathcal{N}(i)$, the directed connectome uses bounded query/key/value maps and a learned edge bias b_{ij}:

$$a_{ij} = \mathrm{softmax}_{j \in \mathcal{N}(i)} \left(\frac{q(h_i)^\top k(h_j)}{T} + b_{ij} \right), \tag{4}$$
$$m_i = \sum_{j \in \mathcal{N}(i)} a_{ij}\, v(h_j). \tag{5}$$

Bounded mode uses damp-only spectral rescaling and unit-ball query/key projection. It never amplifies a vector merely to normalize it. This detail arose from a concrete failure: full RMS normalization amplified naturally small recurrent messages by roughly twenty-fold and produced a long-lived finite-gradient “zombie” before NaNs appeared.

3.3 Private identity, consolidation, and anchors

Cell identity is not shared across the population. Bounded private expression and later private low-rank transition adapters permit specialization within typed tissue. Utility is estimated from held-out gradients on private expression/adapters and installed edges, combined with incoming and downstream demand. Early consolidation multiplied each optimizer delta by a utility-dependent plasticity factor. S1-P4 introduced an immutable anchor θ_A and a proximal restoring step:

$$\theta \leftarrow \theta_A + \frac{\theta_{\text{Adam}} - \theta_A}{1 + \lambda p}, \tag{6}$$

where $p \in [0, 1]$ is protection and $\lambda = 0.01$. Unlike a reduced learning rate, this creates cumulative displacement cost: repeatedly rewriting a mature parameter continues to encounter the anchor.

3.4 Append-only development

Growth appends relay cells without moving existing cells, deleting working edges, or resetting optimizer moments. New state and private adapter-up rows begin at zero. Each new cell receives a sensory-reachable path; dormant dendrite slots provide weak readers into new tissue. A graft ledger records structural changes. The operation itself is now verified, including checkpoint/resume after CUDA growth. Whether growth creates useful capacity is a separate causal question tested by freezing or zeroing the born tissue.

4 Detachable Organs

4.1 Discrete sensor/effector organs

An organ owns its input embedding or continuous sensor, per-input-cell gain and bias, attentive output reader, and decoder. The graph, tissue rules, private cell parameters, and living electrical state belong to the shared organism. Every step explicitly names the active organ. An inactive organ receives no gradient, and its parameter digest can be checked byte-for-byte.

This physical interface boundary repaired an ambiguity in the first procedural-transfer experiment. Remapping one shared symbol interface created contradictory meanings without a mode cue. Independent A and B organs allow the same substrate to be evaluated through distinct ports without demanding that one embedding mean two incompatible things.

4.2 The frozen-LLM language organ

The language graft uses a pretrained decoder-only transformer as replaceable linguistic machinery [2, 8]. For each perceived token:

1. frozen Llama produces a detached contextual hidden feature;
2. a bounded continuous sensor writes a compact representation into memory and drives input tissue;
3. the continuing organism evolves through its directed recurrent anatomy;
4. an attentive reader sees output tissue only;
5. a zero-initialized low-rank decoder emits continuous control tokens; and
6. the frozen final hidden state attends to those controls before the frozen vocabulary head scores the next token.

Let H_t^{out} denote output-tissue state, A_o the attentive organ reader, $B \in \mathbb{R}^{r \times d_{\text{LLM}}}$ a shared low-rank basis, and g the bounded control gain. The effector is

$$C_t = g\,\tanh\big(\tanh(A_o(H_t^{\text{out}}))B\big)\,. \tag{7}$$

The first injection point is immediately before the frozen language head. This avoids retaining a training graph through all 8B transformer parameters and preserves exact zero-control identity. Deeper multi-layer control remains a possible expressivity extension, not an assumption hidden in current results.

The memory recall head is content-addressed in the spirit of Neural Turing Machines and Differentiable Neural Computers [3, 4], but preserves the organism boundary. For query q_t and valid memory cells M,

$$r_t = \gamma\,\tanh\left(W_o \sum_j \text{softmax}_j(q_t^\top k(M_j))\,v(M_j)\right). \tag{8}$$

The recalled vector re-enters sensory/recurrent tissue. It cannot directly reach the language controls or vocabulary head.

What a fluent output would not prove

The frozen language model is already fluent. A compelling sentence is not evidence that the organism contributed. A language result becomes organism evidence only when matched zero-control, reset, wrong-passage, or relevant-tissue lesion arms lose the claimed capability while the backbone remains frozen.

5 Experimental Methodology

5.1 Causal attribution before storytelling

Headline task metrics are paired with interventions on matched examples and cloned states. Reset measures dependence on persistent state. Freeze measures whether a tissue works. Within-tissue shuffle measures differentiation. Identity shuffle measures cell- specific expression. Degree-preserving rewire measures dependence on learned white- matter-like routing. Memory zero, wrong passage, zero control, and organ removal isolate specific paths. None of these insults is used as a robustness objective unless a later recovery experiment explicitly says so.

5.2 Matched controls and calibrated claims

Parameter-matched GRUs and small transformers establish whether a compact task is learnable and quantify the opportunity cost of an evolvable cellular substrate. They do not define the entire project objective: the expected advantages of SOL2 are persistent evolution, structure, growth, and organ integration rather than dominance on small fixed-distribution sequence prediction.

The project has repeatedly corrected over-strong readings of single tests. Examples include distinguishing freeze from shuffle, recognizing that a failed growth treatment does not reject growth, and treating the 25-update wiki gates as routing diagnostics rather than adequate learning horizons. Negative results constrain mechanisms; they do not automatically falsify the architectural family.

5.3 Reproducibility and operational integrity

Result-bearing protocols are written before optimizer updates where practical. Complete checkpoints preserve model parameters, attached organs, optimizer moments, living state, sample cursor, histories, and CPU/CUDA random states. GPU jobs select physical devices by UUID after ordinal selection once placed a run on the wrong card. Dependency-aware campaign manifests are immutable and require both successful exit and declared artifacts. The frozen language backbone is identified by model name and digest but is never copied into organism checkpoints.

6 Results

6.1 From SOL to Fable: establishing a functional substrate

The original SOL character organ used one shared recurrent cell rule on a sparse directed connectome. At 122,306 parameters and 5,000 updates it reached 2.562 final held-out BPC, behind a matched GRU at 2.366 but ahead of the tested transformer at 2.618. Resetting state cost 6.283 BPC and shuffling cells cost 2.265, establishing that the recurrent substrate was not a dead intermediary.

Grok and Fable then isolated what mattered. Additional cross-window credit and early metabolism were neutral or harmful on the closed character objective, while scale and a less destructive readout were useful. Fable F0 matched a schedule-matched GRU within 0.0086 mean BPC across seeds 7, 13, and 21. The head-only control was 1.68 BPC worse, excluding a readout-only explanation. Internal freeze cost approximately 0.51 BPC on average, while within-tissue shuffle cost only about 0.05. The correct interpretation was functional but mostly interchangeable tissue, not unused neurons.

These results closed the elementary feasibility question: a sparse persistent cellular field can learn a continuous character stream and use its internal state. They did not establish the differentiation, continual retention, or growth properties sought by the project, motivating SOL2.

6.2 S0-R: replicated stable differentiation

SOL2 compared identity-conditioned bounded and unbounded operators over seeds 7, 13, and 21. All six 2,000-update arms completed with zero rejected updates. Bounded operators paid a mean capability cost of 0.0961 BPC (2.8972 versus 2.8010), but produced far stronger causal structural dependence:

Treatment	Held-out BPC	Internal freeze Δ	State shuffle Δ	Identity shuffle Δ	Topology rewire Δ
Identity, bound 4	2.8972	+3.4068	+0.1220	+0.0728	+2.3069
Identity, unbounded	2.8010	+3.0808	+0.0136	+0.0038	+0.4863

Table 1: *Three-seed mean S0-R results. Lower BPC is better; positive intervention deltas mean the learned capability depended on the intervened substrate.*

Mechanistic evidence The bounded identity treatment exhibited replicated positive state, identity, topology, and freeze effects in every seed. This is the strongest evidence that SOL2 learned a differentiated cellular computation rather than merely passing signal through an undifferentiated field. The treatment was promoted despite its modest closed-task opportunity cost.

6.3 S1-P: procedural acquisition and transfer

The procedural benchmark generates fresh programs from four transformations and scores only the exact final value. Memorizing facts cannot solve it. The first 310,390-parameter creature reached only 31.9% length-four accuracy at the initial curriculum fork, then reached 60.4% after 2,000 more A updates. On a new B interface, the acquired organism reached 54.4% while an identically trained B-from-scratch organism remained at 11.2%. The acquired organism held a preregistered +12.08-point early advantage.

Freezing compute, relay, or all internal tissue reduced A performance by roughly 47–49 points, and topology rewiring cost 46.7 points. This showed that the procedural result used both grey-like computation and white-like routing. Organ-only B reuse remained strong at shallow lengths but failed at length four, indicating that deeper transfer required plasticity inside the integrated substrate.

6.4 S1-P2: true organ attachment, forgetting, and recovery

A larger/deeper 1.224M-parameter organism mastered A at update 5,000 with consecutive 97.66% and 98.76% length-four checks. A new independently parameterized B sensor/ effector organ was then attached. The mature substrate learned B much faster than a fresh substrate: it reached 50% at update 800 versus 1,800 for scratch, and 80% at 1,000 versus 2,000. Terminal B accuracy was 95.77% versus 84.83% for scratch.

The unchanged A organ did not preserve A behavior: A fell to 12.43% while matched A-control remained at 99.35%. Compute, relay, internal, and topology lesions cost 72–86 points on B, excluding an organ-only bypass. When B was removed and the organism was retrained on A, A returned to 98.96%, but the byte-identical reattached B initially scored 23.57%. B then recovered to 79.36% in 25 updates and 97.40% in 100.

Interpretation of S1-P2

Mechanistic evidence The substrate is highly reusable and rapidly rewritable; prior procedural development materially accelerates learning through a new organ. The same evidence shows that the current organism stores only one deep active procedural mode at a time. Organ parameter isolation is solved; simultaneous internal mode retention is not.

6.5 S1-P3 through S1-P4: the stability–plasticity frontier

S1-P3 enlarged the organism to 400 cells and protected mature structure according to first-order utility. Plastic adaptation produced A/B accuracies of 13.48%/97.46%. Measured consolidation produced 86.98%/33.72%, improving the worst capability by 20.25 points, but shuffled protection was equally effective. The treatment validated heterogeneous slowing and reserve drift, not the utility allocator.

S1-P3b added rank-four private transition residuals and directional reserve edges. Resetting private adapters cost roughly 47–54 B points, proving that the new cell-local computation was recruited. Yet B still learned mainly by reusing and overwriting high-A-utility cells; new reserve grafts were nearly non-causal, and no arm reached simultaneous 80/80.

S1-P4 replaced per-step slowing with cumulative anchors and made growth conditional on measured plasticity pressure. The seed-7 result was:

Branch	A	B	$\min(A, B)$	Grown cells
Plastic	14.06%	93.10%	14.06%	0
Uniform anchor	50.91%	57.29%	50.91%	0
Measured anchor	64.00%	60.09%	60.09%	0
Developmental	72.40%	57.81%	57.81%	32

Table 2: *S1-P4 terminal exact length-four accuracy. Measured anchors produced the best worst-organ frontier. The developmental branch grew successfully but did not recruit the new cells into the B computation.*

Measured anchors improved $\min(A, B)$ by 9.18 points over uniform anchors, narrowly missing the frozen 10-point attribution gate. The developmental arm triggered twice and appended 32 relay cells without disrupting the organism, but freezing those cells cost only 0.20 B points and zeroing their adapters cost 0.33. Thus pressure detection and append-only morphogenesis work mechanically; making new tissue the preferred route for new capability remains open.

6.6 L0: integrating a frozen language model

The first production graft used NousResearch/Meta-Llama-3-8B-Instruct in BF16 on a physical RTX 4090. The 8.03B-parameter backbone had zero trainable parameters, zero gradient tensors, and exactly zero native-logit error through the adapter. The 2.36M- parameter organism/graft began with exact-zero controls and received a nonzero control- decoder gradient. Peak reserved memory was 16.26 GB, and a graft forward/backward step took 0.327 seconds.

Established This is a genuine organ boundary: the pretrained language system can be attached without fine-tuning, the new learned path resides in the cellular organism and graft, and zero control exactly recovers the frozen model. It is not yet evidence of organism-level conversation.

6.7 L0-C1: held-out wiki memory

The wiki experiment asks whether the organism can absorb a passage, survive erasure of the Llama token/KV workspace, and later steer the same frozen model to answer. Source families are disjoint, facts are baseline-screened, and matched arms include no exposure, zero control, reset, wrong passage, memory lesion, internal lesion, and paraphrase. Counterfactual paired bindings present the identical question with incompatible episodic answers, preventing a static Llama prior from satisfying both branches.

The experiment series has been diagnostically productive:

Run	Intervention	What improved	Remaining failure
C1a	Initial 50 updates	Nonzero language steering	Query overwrote all 16 memory slots; causal arms equal
C1b	Query write gate, paired branches, 96 slots	Passage state retained; held-out normal 62.5%	No-exposure/reset/wrong passage also 62.5%; static steering
C1c	Differentiable writes and content recall	Memory presence reached recurrent route	Paired control separation 6.99×10^{-6}
C1d	Bidirectional binding margin	Correct causal objective	Signal diluted by long passage; paired accuracy 10%
C1e	Compact binding curriculum	First nonzero label separation; control separation 4.8× C1d	Effector gradient exceeded recall by $> 500\times$
C1f	Recall gain and optimizer rebalance	Strong memory (0.289 RMS) and internal (0.00729) branch separation localized	Output 0.000616; control below BF16 resolution
C1g	Bounded control gain 64	Update-2 label separation 0.0884; numerical no-op repaired	At update 25, content-dependent control not learned

Table 3: *L0-C1 is a sequence of localized mechanism tests, not seven independent capability claims.*

At the C1g update-25 diagnostic, held-out normal accuracy was 37.5%, equal to no exposure, reset, wrong passage, memory lesion, and internal lesion. However, matched incompatible branches remained strongly separated in memory (0.2816 RMS) and measurably separated in internal tissue (0.00551), collapsing at output tissue (0.000393). Gain 64 made early Llama logit differences numerically resolvable but did not align them with the correct branch within the short horizon.

Twenty-five updates cover only about two visits to each of 12 meta-training questions. It is therefore an architectural smoke gate, not a fair training horizon for jointly learning memory addressing, cellular transport, output expression, and the frozen Llama head's control geometry. The next experiment is an explicitly exploratory continuation of the exact C1g checkpoint to 300 updates. Because that horizon was selected after observing update 25, any result will be reported as hypothesis-generating rather than as a retroactive pass of the preregistered short gate.

7 What Has Been Achieved

Validated achievements as of 4 August 2026

1. **Established** A persistent, checkpointable typed cellular organism with bounded state, sparse directed anatomy, private cell computation, dedicated organ sinks, and append-only growth has been implemented and tested.
2. **Mechanistic evidence** Cellular state, compute tissue, relay tissue, private identity, and learned topology have each been shown causally necessary in replicated or matched interventions.
3. **Mechanistic evidence** A mastered organism learns a deep procedure through a new organ substantially faster than a fresh substrate on the same examples.
4. **Established** Physical organs can be attached, removed, kept byte-identical, reattached, and recovered without rebuilding the organism.
5. **Mechanistic evidence** Private low-rank cell transitions are recruitable and jointly causal; cellular differentiation is no longer limited to one shared parameter state.
6. **Mechanistic evidence** Cumulative anchors substantially improve simultaneous retention and adaptation compared with unconstrained plasticity.
7. **Established** Plasticity pressure can trigger append-only growth without corrupting old anatomy, state, or optimizer history, including exact CUDA resume.
8. **Established** A frozen 8B Llama model can function as a detachable continuous language organ with exact baseline parity and no backbone gradients.
9. **Diagnostic** Wiki bindings survive in memory and propagate into internal tissue after LLM context erasure; the current bottleneck is training their content-dependent expression through output tissue and the language head.

These achievements are meaningful but narrower than the dream. The current organism is not yet an autonomous conversational creature, has not demonstrated indefinitely stable lifelong learning, and has not shown that grown neurons improve capability. The value of the program is that those statements are now localized engineering and scientific questions rather than undifferentiated speculation.

8 Open Limitations

8.1 One deep mode at a time

The clearest capability limitation is destructive interference inside the shared substrate. Independent organs preserve their own parameters, but whole-organism learning rewrites the procedure expressed by shared tissues. Anchors improve the frontier without solving simultaneous mastery. A low-rank organ-conditioned dynamic mode state is the most direct next mechanism: per-lane coefficients should modulate cell-specific target and timescale bases, allowing overlapping circuits to express different configurations.

8.2 Growth is mechanically correct but economically unattractive

Born relay cells are reachable and non-destructive, yet trained branches continue to modify mature high-utility circuitry. Growth must change the optimization landscape so new tissue is easier to recruit than anchored tissue is to overwrite. Candidate methods include stronger cumulative costs, local auxiliary competence at new cells, demand- matched reader installation, and longer post-growth training before attribution.

8.3 The language effector is shallow

Current controls enter immediately before the frozen vocabulary head. This makes the zero-control invariant and training memory tractable, but asks the organism to learn the geometry of a large frozen embedding head through a low-rank final residual. Deeper zero-residual injection, per-layer adapters, or a small learned interface model may be more expressive, provided the frozen backbone remains frozen and the path continues to originate at output tissue.

8.4 Memory training is underpowered

The current wiki curriculum is tiny, and the diagnostic gates intentionally stopped most treatments after 25 updates. The internal signal demonstrates routing feasibility, not a learned general memory algorithm. Longer training, more varied paired bindings, distractor restoration, delay sweeps, and later short-answer evaluation are required.

8.5 True continual and metabolic learning remain open

Current optimization truncates gradient history at weight-version boundaries. Eligibility traces, reverse credit, and randomized causal structural traffic were explored extensively in SOL, but their capability benefit was inconsistent. Liquid time constants and closed- form continuous dynamics suggest a principled route to irregular organ latencies [5, 6]; e-prop supplies a reference for local event traces meeting delayed learning signals [7]. These mechanisms should return only behind tasks that require them, not as biological ornament.

9 Research Agenda

1. **Give L0-C1g enough learning time.** Continue the exact checkpoint to 300 updates, evaluate every 50, preserve paired counterfactuals, and judge the full trajectory. If logit separation reappears and becomes target-aligned, extend the curriculum from compact bindings toward natural wiki distractors.
2. **Localize output-tissue learning if time alone fails.** Compare an anatomy-neutral output-state contrastive reward, additional microsteps, and a learned relay-to-output developmental path. Do not add a direct memory-to-LLM bypass.
3. **Implement organ-conditioned dynamic modes.** Train interleaved A/B switches so the same mature anatomy can express more than one deep procedure. Only after a useful active mode exists should DNC-like memory store and retrieve compact mode coefficients or transition links.
4. **Make growth recruitable.** Use anchor pressure as the growth trigger but make born cells the low-cost path for new computation. Require a causal removal penalty before claiming development.
5. **Adopt delta-aware liquid timescales.** Replace implicit unit time with $\alpha = 1 - \exp(-\Delta t/\tau_i)$, using private baseline timescales and bounded contextual modulation. Test irregular sensor and organ latency, not character BPC alone.

6. **Scale across organism size and seeds.** The prepared campaign spans 400–3,200 cells and seeds 7, 13, and 21 with mastery-gated dependencies. Larger compute should strengthen inference, not merely multiply exploratory runs.

7. **Build a living visualization.** Persist stable neuron identities, tissue roles, activation, utility, edges, birth events, and organ controls so a user can see learning and growth without confusing activity with importance.

8. **Return metabolism carefully.** Reward reducible surprise, enforce conservation across every operator, include satiety, and compare against a no-economy organism before coupling survival to human attention.

10 Conclusion

The project has crossed the line from an evocative metaphor to an experimentally tractable organism architecture. SOL2 is not a conventional language model in a cellular costume: its state persists, its typed tissues and directed anatomy are causal, its organs are physically separable, its substrate transfers learned procedural structure, and its developmental operations preserve a living process. At the same time, the strongest failures are informative. The organism remains prone to rewriting one deep mode with another; grown cells are not yet preferred by learning; and retained wiki information has not yet become reliable content-dependent language control.

The conceptual route remains feasible. A frozen LLM can plausibly serve as one organ inside a larger adaptive substrate, while cellular memory, structure, timescale, growth, and cross-organ state supply capabilities absent from the language model alone. The next phase should resist both premature scale and premature rejection: train the mechanisms long enough to learn, insist on causal controls, and let each negative result identify the next anatomical or learning bottleneck.

A Evidence and Artifact Index

Topic	Repository evidence
Project goals and safety	`PROJECT.md`; `ARCHITECTURE.md`
SOL character organism	`sol/README.md`; `sol/runs/S0-COMPARISON.md`
Grok/Fable synthesis	`grok/findings.md`; `fable/experiments/f0-concat-close-out.md`; `fable/experiments/f2-report.md`
SOL2 kernel	`sol2/README.md`; `sol2/model.md`; `sol2/tissues.md`; `sol2/graph.md`
Stable differentiation	`sol2/experiments/s0r-stable-differentiation.md`; bead `DMON-1-prt`
Procedural transfer	`sol2/experiments/s1-procedural-transfer.md`
Acquisition scaling	`sol2/experiments/s1p2-acquisition-scaling.md`
Organ attachment	`sol2/experiments/s1p2-organ-attachment.md`; `s1p2-organ-attachment-result.json`
Consolidation and reserve	`sol2/experiments/s1p3-consolidated-reserve.md`; `s1p3b-private-transition-reserve.md`
Anchored development	`sol2/experiments/s1p4-anchored-development.md`; `s1p4-anchored-development-result.json`
Frozen language organ	`sol2/experiments/l0-living-language-organ.md`; `l0-llama3-8b-smoke-result.json`
Wiki memory series	`sol2/experiments/l0c1-wiki-memory.md`; `l0c1*-result.json`

B Selected External References

References

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